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RESEARCH ARTICLE

Stackelberg Game Based on Trajectory Prediction for Lane Change in Mixed Traffic

BAICHUAN SHI[1], LI ZHAI[1], (Member, IEEE), AND CHANG LIU[2]
[1]School of Mechanical Engineering and Vehicle Engineering, Beijing Institute of Technology, Beijing 100081, China
[2]China Automotive Engineering Research Institute Intelligent Connected Technology Company Ltd., Suzhou 215100, China
Corresponding author: Li Zhai (zhaili@126.bit.edu.cn)

ABSTRACT To address lane-changing conflicts between intelligent and human-driven vehicles in mixed traffic environments, this study proposes a Stackelberg game-based decision-making method for autonomous vehicles. A Stackelberg game framework is established between autonomous vehicles (leaders) and target-lane human-driven vehicles (followers) in three typical scenarios. The method develops a utility function for human-driven vehicles incorporating driving styles and safety-comfort-efficiency factors, with a corresponding cost function for autonomous vehicles. An improved Stackelberg game model integrates trajectory prediction of human-driven vehicles, while a bi-level optimization algorithm combining model predictive control and genetic algorithms jointly optimizes acceleration sequences and lane-change timing. Simulations demonstrate 60% reduction in heading angle variation and 67.59% decrease in yaw rate compared to non-game strategies, confirming enhanced safety, comfort, and efficiency of the proposed method.

INDEX TERMS Stackelberg game, lane-changing decision-making, mixed traffic environments, trajectory prediction, bi-level optimization.

I. INTRODUCTION

With the rapid development of Intelligent and Connected Vehicle (ICV) technology, the safety of autonomous vehicles (AVs) in complex traffic environments faces increasingly stringent requirements [1]. Autonomous lane-changing, as one of the core functionalities of AVs, directly impacts driving safety and traffic efficiency through the rationality of its decision-making. However, in mixed traffic scenarios, AVs must dynamically interact with human-driven vehicles (HVs), where diverse driving styles (e.g., conservative yielding or aggressive confrontation) among HVs can easily lead to lane-changing conflicts. Achieving safe, comfortable, and efficient lane-changing through cooperative decision-making has become a critical challenge for intelligent transportation systems [2], [3].

Existing lane-changing decision-making methods can be categorized into three types: rule-based, data-driven, and game-theoretic approaches. Rule-based methods trigger lane-changing behaviors through predefined conditions. Early studies, such as the Gipps model [4], defined safety distance thresholds, while subsequent works improved generalization by incorporating driver behavior models [5] and dynamic safety assessment mechanisms [6]. However, such methods often fail in complex interaction scenarios due to rigid rule constraints.

Data-driven methods optimize decisions via machine learning. Early research employed support vector machines [7] and Markov decision processes [8], while recent studies focus on hierarchical reinforcement learning [9] and rule-constrained deep reinforcement learning [10], with the latter enhancing adaptability to complex environments through reward function design. Nevertheless, these methods rely on large-scale training data and lack robustness guarantees in real-world verification scenarios [2], [11].

Game-theoretic methods, which model multi-agent interactions to balance safety and efficiency, have gradually emerged as a research hotspot. Early studies (e.g., [12]) utilized Nash equilibrium to predict driver behavior under the assumption that participants maximize their own interests. Recently, dynamic game frameworks [13] and Stackelberg games [14] have been introduced into lane-changing

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scenarios. Non-Stackelberg games (e.g., Nash games) assume synchronous decision-making among participants, which overlooks the delayed response and decision order of human drivers (HVs) in human-machine mixed driving environments, leading to strategic conflicts. In contrast, Stackelberg games employ a "leader-follower" architecture (e.g., AVs as leaders making prioritized decisions, and HVs as followers responding accordingly) to optimize strategies, thereby improving overall traffic flow. Simulations indicate that Stackelberg games reduce lane-changing conflict rates by over 30% [15], [16], better aligning with the interaction demands of mixed traffic scenarios.

Stackelberg game-based decision frameworks demonstrate significant advantages in mixed traffic scenarios. In [15], a collaborative decision-making framework was proposed for multi-lane merging zones, defining connected autonomous vehicles as leaders and HVs as followers. Building on this, Hang et al. [16] systematically incorporated quantified driving styles (aggressive, moderate, conservative) and social rule modeling (e.g., yielding behavior) into the game framework. Experiments showed that compared to Nash equilibrium, Stackelberg games reduce comprehensive decision costs (including safety risks and energy consumption) by 23.5% under typical driving styles. For forced lane-changing scenarios, Huang et al. [17] developed a decision model based on dynamic Stackelberg games, introducing a Relative Style Index to quantify driver safety threshold differences, achieving a 32% improvement in lane-changing success rates. To further adapt to dynamic traffic flows, Yao and Jin [18] proposed a multi-lane game model that dynamically selects game participants based on real-time traffic states (e.g., traffic density, speed distribution) and designs priority weight allocation mechanisms, addressing the rigidity caused by fixed participant assumptions in traditional models. Additionally, Zhang et al. [19] innovatively combined Stackelberg games with a hierarchical MPC architecture: the upper layer optimizes target vehicle decision weights via inverse deduction, while the lower layer adjusts control inputs using real-time feedback. This approach reduces computational latency to below 50 ms while ensuring interaction safety, significantly enhancing system adaptability to unknown opponent strategies.

Despite progress in leader-follower game-based lane-changing decision-making, the following issues remain unresolved:

Homogeneous Agent Modeling: Existing models do not distinguish the heterogeneity between HVs and AVs, employing identical utility functions and neglecting the goal divergence where HVs prioritize efficiency while AVs emphasize safety [20].

Insufficient Dynamic Interaction Modeling: Most studies fail to couple driving style diversity (e.g., aggressive HVs accelerating to occupy lanes or conservative HVs decelerating to yield) with the influence of surrounding vehicle trajectory predictions, resulting in insufficient robustness in mixed traffic scenarios [21].

Oversimplified Control Variables: Current methods optimize only lane-changing timing while neglecting acceleration sequence planning, leading to peak jerk values exceeding 2.5 m/s^3 during lane changes, significantly compromising ride comfort [22].

The contributions of this paper are as follows:

(1) A driving style-trajectory prediction-integrated leader-follower game architecture is proposed, designing differentiated comprehensive utility functions (incorporating safety, positional advantage, comfort, and efficiency) for HVs and constructing a bi-layer cost function for AVs that jointly optimizes acceleration sequences and lane-changing timing.

(2) A bi-layer optimization framework based on MPC and genetic algorithms (GA) is developed to enable real-time dynamic planning of lane-changing strategies.

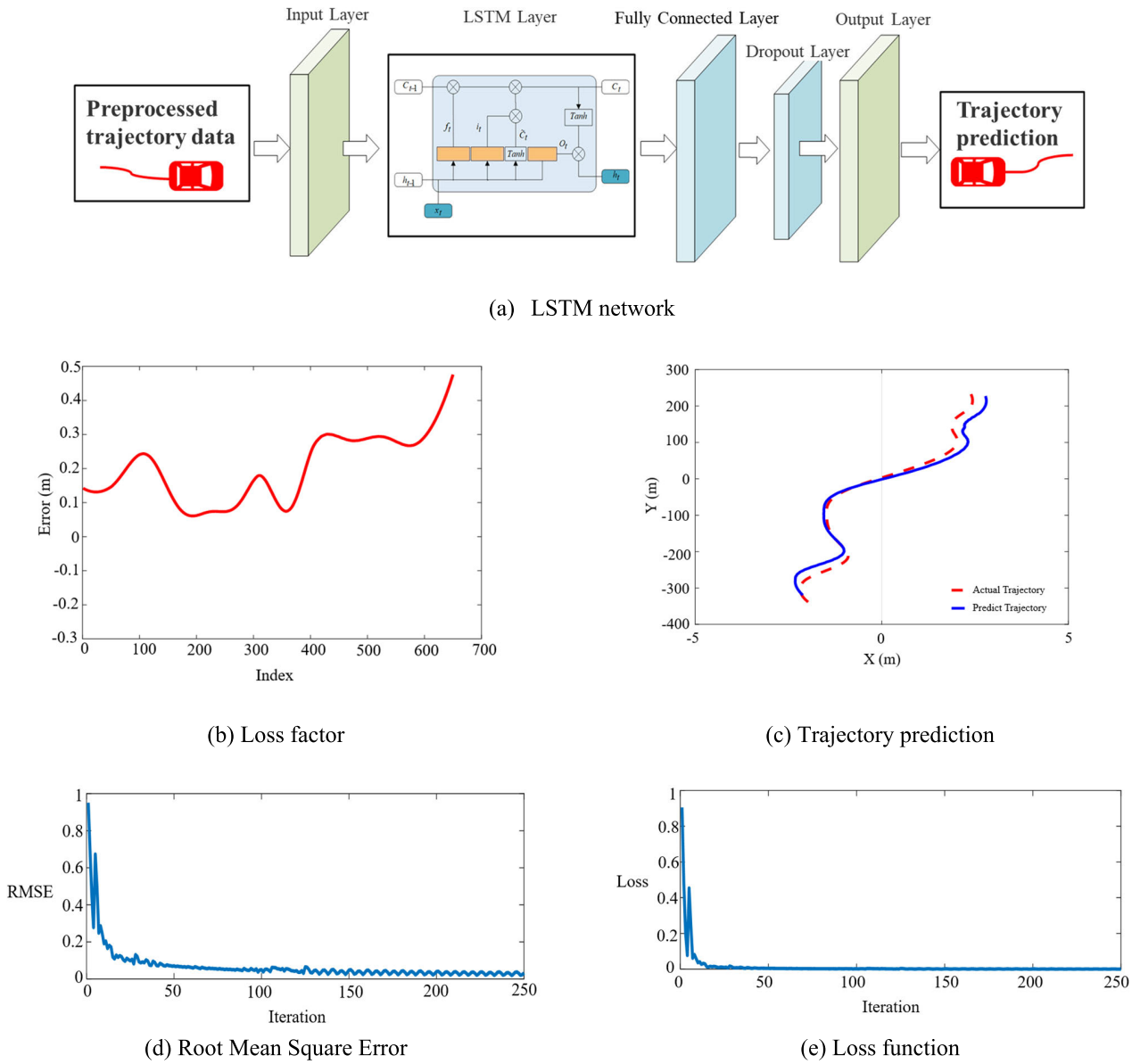
(3) Through co-simulation using Prescan-Carsim-Simulink, the proposed method reduces variations in heading angle and yaw rate, significantly improving lane-changing smoothness.

II. TRAJECTORY PREDICTION

We investigate three distinct categories of trajectory data: lane-keeping trajectories, left lane-changing trajectories, and right lane-changing trajectories. A K-means-based clustering algorithm was implemented, with the initial cluster centers set to 3, to partition the lane-changing trajectory dataset into three clusters. As proposed in [23], driver style categorization validity was determined through statistical analysis of acceleration/deceleration patterns. The evaluation methodology for sudden acceleration/deceleration frequency through threshold-based temporal proportion calculations was proposed. Specifically, the proportion of time frames with positive acceleration below predefined thresholds (indicating sudden acceleration) and negative acceleration exceeding specific thresholds (denoting sudden deceleration) were computed across complete driving cycles. The results reveal that aggressive drivers exhibit significantly higher occurrences of both sudden acceleration and deceleration compared to other driver types, aligning with established driving behavioral character.

A deep neural network architecture comprising an input layer, an LSTM layer, a Fully-connected layer, and a Dropout layer was implemented to train trajectory prediction models using preprocessed trajectory sequences. The LSTM-based trajectory prediction was subsequently conducted on left/right lane-changing vehicle data extracted from the NGSIM dataset.

As illustrated in Figure 1(d)(e), both the loss function and Root Mean Square Error (RMSE) values converged below 0.1 during training iterations, asymptotically approaching zero. Specifically, Figure 1(b) demonstrates lateral prediction errors not exceeding 0.1 meters for left lane-changing trajectories, thereby validating the effectiveness of the proposed prediction framework.



(a) LSTM network

(b) Loss factor

(c) Trajectory prediction

(d) Root Mean Square Error

(e) Loss function

FIGURE 1. LSTM network and prediction results.

III. MODEL

A. LANE-CHANGING SCENARIOS

Lane-changing scenarios are typically categorized into voluntary and mandatory types based on vehicle intent. Three representative lane-changing scenarios were established as shown in Figure 2.

Figure 2(a) depicts a bidirectional dual-lane voluntary lane-changing scenario. The red vehicle represents the Ego Vehicle (EV) with autonomous driving capability, the black vehicle indicates the preceding vehicle FV2 in original Lane 2, while the blue vehicle denotes human-driven rear vehicle RV1 in target Lane 1. In this configuration, EV initiates lane-changing to overtake FV2 due to unsatisfactory driving speed. Figure 2(b) presents a tri-lane voluntary lane-changing scenario where EV initially travels in Lane 2 flanked by vehicles in both adjacent lanes (Lane 1 and Lane 3). Compared to the dual-lane scenario in Figure 2(a), EV faces three behavioral options: lane-keeping car-following, left lane-changing, or right lane-changing. The decision-making process requires comprehensive evaluation of both rear vehicles (RV1/RV3) and preceding vehicles (FV1/FV3) in target lanes, with optimal lane-changing strategies determined through a Stackelberg game-theoretic decision-making algorithm. Figure 2(c) illustrates a mandatory lane-changing scenario where preceding vehicle FV3 from Lane 3 abruptly merges ahead of EV in Lane 2, compelling EV to execute emergency lane-changing. All three scenarios fundamentally reflect interactive game-theoretic

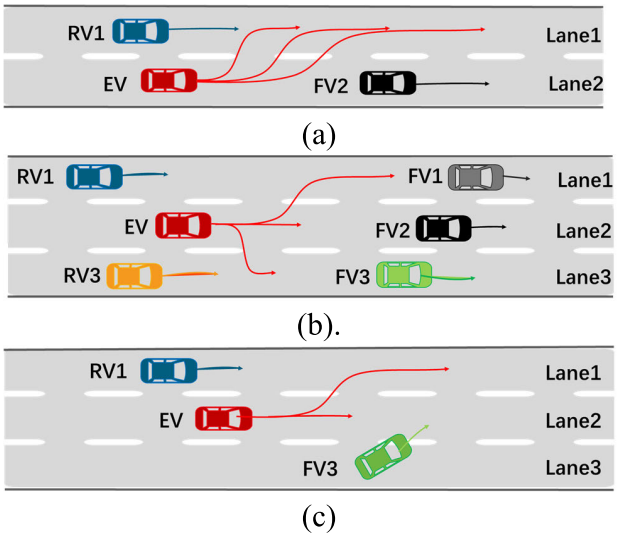


FIGURE 2. Typical lane-change scenarios.

processes between autonomous vehicles and human-driven vehicles in target lanes.

B. STACKELBERG GAME MODEL

During lane-changing maneuvers, the autonomous ego vehicle (EV) engages in game interactions with human-driven rear vehicles in the target lane while maintaining safe distances from preceding vehicles in both the current and target lanes to ensure safety. As illustrated in Figure 3, the EV selects target lanes based on collision risk assessment results.When the target lane's rear vehicle is human-driven (subsequently referred to as HV), the autonomous ego vehicle (EV) assumes the leader role in a Stackelberg Game framework, with the HV (RV1) acting as the follower. During the interaction, the EV first activates turn signals to communicate its lane-changing intent to the HV, which responds by either decelerating (conservative HVs) to accommodate the maneuver or maintaining/accelerating speed (aggressive HVs) to force the EV to abort. Leveraging advanced perception systems, the EV predicts surrounding vehicle trajectories, evaluates safe lane-changing distances, and classifies HV driving styles, enabling real-time adjustments to its speed and lane-changing timing to ensure safe and smooth transitions.

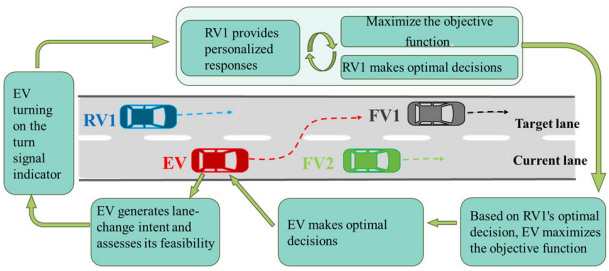


FIGURE 3. Stackelberg game-based autonomous vehicle lane change.

The decision-making process based on the stackelberg game can be described as follows:

$$\gamma^{RV1} = \underset{s^{RV1}\in S^{RV1}}{arg\,min}\left(\underset{s^{RV1}\in S^{RV1}}{max}\ J^{RV1}\left(state, s^{EV}, s^{RV1}\right)\right)$$
$$\gamma^{EV} = \underset{s^{EV}\in S^{EV}}{arg\,min}\left(\underset{s^{RV}\in S^{RV1*\prime}(s)}{max}\ J^{EV}\left(state, s^{EV}, s^{RV1}\right)\right)$$
$$S^{RV1*} = \left\{\begin{array}{l} s^{RV1\prime}\in S^{RV} : J^{RV}\left(state, s^{EV}, s^{RV1\prime}\right) \geq \\ J^{RV}\left(state, s^{EV}, s^{RV1}\right), \forall s^{RV1}\in S^{RV1} \end{array}\right\} \quad (1)$$

s^{EV} and s^{RV1} represent the decision actions of the game participants, EV and RV1, while s^{EV} and s^{RV1} denote the corresponding decision sets for both. $s^{RV1\prime}$ represents the optimal strategy of RV1. S^{RV1*} represents the set of optimal strategies ofThe payoff functions for EV and RV1 are denoted as $J^{EV}\left(state, s^{EV}, s^{RV1}\right)$ and $J^{RV1}\left(state, s^{RV1}, s^{EV}\right)$, respectively, with *state* representing the vehicle's state information at the current moment. γ^{EV} and γ^{RV1} are the equilibrium solutions obtained through the game model, representing the optimal decisions for the game participants at the current time.

IV. DECISION MAKING BASED ON STACKELBERG GAME

A. LONGITUDINAL CONTROL BEHAVIOR MODEL

Traditional vehicle acceleration models typically assume that acceleration is constant or piecewise constant, ignoring the continuous variation of actual acceleration with speed. To better reflect the actual driving behavior of human drivers and the vehicle's acceleration performance, an improved linear-decreasing acceleration model [24] is used. In this model, the vehicle's acceleration decreases linearly as speed increases, and it is represented as:

$$a_x = \beta\left(v_{target} - v_{init}\right)e^{-\beta t} \quad (2)$$

a_x represents the longitudinal acceleration, v_{target} is the target speed, v_{init} is the initial speed at the starting moment, β is the slope of the acceleration curve.

Based on the acceleration model in Eq. (2), integrating it yields the expressions for speed and displacement:

$$v = v_{target} - \left(v_{target} - v_{init}\right)e^{-\beta t} \quad (3)$$
$$x = x_{init} + v_{target}t - \left(v_{target} - v_{init}\right)\left(1 - e^{-\beta t}\right)/\beta \quad (4)$$

In this equation, v represents the current speed of the vehicle, x represents the current longitudinal position, and x_{init} is the longitudinal position at the initial moment.

Based on the longitudinal maneuvering behavior model of human drivers established in equations (2) to (4), once the parameters v_{init}, x_{init} and v_{target} are known, the motion state and behavior of a human-driven vehicle can be calculated. Different driving styles are defined based on these parameters. When the autonomous vehicle (EV) intends to change lanes, a normal or conservative driver in the target lane usually expects to decelerate or maintain their speed while keeping a larger distance from the EV to avoid collision and ensure safety. In contrast, an aggressive driver will accelerate

to a higher target speed in the target lane to prevent the EV from inserting into the lane. The decision variable for human-driven vehicles is defined as $S_{HV} = \{v_{target}\}$.

B. INTEGRATED GAME PAYOFF FUNCTION

In the interaction between the autonomous vehicle (EV) and the human-driven vehicle (RV) during lane changes, factors such as safety, positional advantage, comfort, and traffic efficiency are considered. A payoff function for the lane-change decision of the human-driven vehicle is established, incorporating the influence of different driving styles. This leads to the formulation of an integrated payoff function for the lane-changing game between the autonomous vehicle and the human-driven vehicle.

The driver's judgment of the safe distance is primarily influenced by two factors: the Time to Collision (TTC) and the Headway Distance.

$$TTC(t) = \begin{cases} \dfrac{x_{EV}(t) - x_{HV}(t)}{v_{HV}(t)}, \\ \qquad x_{EV}(t) > x_{HV}(t) \&\& v_{HV}(t) > v_{EV}(t) \\ +\infty, \\ \qquad x_{EV}(t) > x_{HV}(t) \&\& v_{HV}(t) < v_{EV}(t) \\ \dfrac{x_{HV}(t) - x_{EV}(t)}{v_{EV}(t)}, \\ \qquad x_{EV}(t) < x_{HV}(t) \&\& v_{HV}(t) < v_{EV}(t) \\ +\infty, \\ \qquad x_{EV}(t) < x_{HV}(t) \&\& v_{HV}(t) > v_{EV}(t) \end{cases} \tag{5}$$

$$Th(t) = \begin{cases} \dfrac{x_{EV}(t) - x_{HV}(t)}{v_{HV}(t)}, & x_{EV}(t) > x_{HV}(t) \\ \dfrac{x_{HV}(t) - x_{EV}(t)}{v_{EV}(t)}, & x_{EV}(t) < x_{HV}(t) \end{cases} \tag{6}$$

In the equation, v_{HV} and v_{EV} represent the speeds of the human-driven vehicle (HV) and the autonomous vehicle (EV) at any given moment, respectively, while x_{HV} and x_{EV} represent the longitudinal positions of HV and EV at any given moment.

The closer the vehicle's reaction time is to Th, the safer the vehicle is. Therefore, the safety payoff function J^{HV}_{safe} is defined as follows:

$$J^{HV}_{safe} = \frac{Th(tlc)}{\tau} + \frac{2a^* \times TTC(tlc)}{v^{HV}_x(tlc) + v^{EV}_x(tlc)} \tag{7}$$

The reaction time, denoted as τ, includes the vehicle's driving system reaction time τ_1, typically taken as $\tau_1 = 0.05s$, and the driver's reaction time τ_2, typically taken as $\tau 2 = 0.6s$. tlc represents the time required for the lane change, typically taken as $tlc = 3s$. $v^{HV}_x(tlc)$ and $v^{EV}_x(tlc)$ represent the longitudinal speeds of the human-driven vehicle (HV) and the autonomous vehicle (EV) at the end of the lane change, respectively. $Th(tlc)$ represents the headway distance at the end of the lane change, while $TTC(tlc)$ represents the time to collision at the end of the lane change. Lastly, $a^* = 6m^2/s$ denotes the acceleration during braking.

The relative position between the HV and EV at the end of the lane change, expressed as the ratio of the desired headway distance, is used to represent the positional advantage payoff function J^{HV}_{space} for the HV.

$$J^{HV}_{space} = \begin{cases} -1, & \Delta x \le -\Delta x_{lim} \\ \dfrac{\Delta x}{\Delta x_{lim}}, & -\Delta x_{lim} < \Delta x < \Delta x_{\lim} \\ 1, & \Delta x \ge \Delta x_{lim} \end{cases} \tag{8}$$

Δx_{lim} is a constant positively correlated with the reaction time.

The comfort payoff function J^{HV}_{com} for the HV is measured by acceleration.

$$J^{HV}_{com} = -\max_{t\in[0,tlc]} \left| \frac{a_{HV}(t)}{a_{max}} \right| \tag{9}$$

a_{HV} represents the acceleration chosen by the human driver, and a_{max} is the maximum allowable acceleration.

Finally, the traffic efficiency payoff J^{HV}_{eff} is characterized by the deviation between the actual speed v_{HV} of the HV and the desired speed v_{target}.

$$J^{HV}_{eff} = -\max_{t\in[0,tlc]} \left| \frac{v_{HV}(t) - v_{target}}{\Delta v_{max}} \right| \tag{10}$$

where Δv_{max} is the maximum speed deviation threshold.

Based on equations (6) to (9), and combining the above payoff functions, the integrated payoff function J_{HV} for human-driven vehicles with different driving styles is established, represented as:

$$J_{HV} = \omega^{HV}_{safe} J^{HV}_{safe} + \omega^{HV}_{space} J^{HV}_{space} + \omega^{HV}_{com} J^{HV}_{com} + \omega^{EV}_{eff} J^{HV}_{\text{eff}} \tag{11}$$

In the equation, ω^{HV}_{safe}, ω^{HV}_{space}, ω^{HV}_{com}, and ω^{EV}_{eff} are the weight factors for the various payoff functions in the game, assigned different values based on different driving styles. The specific values are set according to the particular lane-changing scenario. For example, a conservative driver typically slows down to maintain a larger gap for safety. A typical driver focuses more on maintaining driving continuity and comfort. An aggressive driver accelerates to occupy a better road position and prevent other vehicles from merging.

C. COST FUNCTION FOR AUTONOMOUS VEHICLES

The driving safety cost for the intelligent autonomous vehicle includes both longitudinal and lateral safety costs, which are related to the safety distance between the vehicle and the front vehicle (FV), as well as the safety distance between the vehicle and the vehicle in the adjacent lane (HV). The driving safety cost function C^{EV}_{safe} is defined as follows:

$$C^{EV}_{safe} = \left(\sigma^2 - 1\right)^2 C^{EV}_{safe-\log} + \sigma^2 C^{EV}_{safe-lat} \tag{12}$$

where $C^{EV}_{safe-log}$ and $C^{EV}_{safe-lat}$ represent the longitudinal and lateral driving safety costs, respectively. $\sigma = \{-1, 0, 1\}$

represents left lane change, maintaining the original lane, and right lane change.

The longitudinal driving cost is related to the relative distance and relative speed with the front vehicle (FV), and is expressed as:

$$\begin{aligned} C_{safe_log} &= k_{v_log}\lambda_{log}\left(\Delta v_{x_log}\right)^2 + k_{s_log}/\left[\left(\Delta s_{x_log}\right)^2+\varepsilon\right] \\ \Delta v_{x_log} &= v_x^{FV} - v_x^{EV} \\ \Delta s_{x_log} &= \sqrt{\left(x^{FV}-x^{EV}\right)^2+\left(y^{FV}-y^{EV}\right)^2} - l_v \\ \lambda_{log} &= \begin{cases} 1, & \Delta v_{x_log} < 0 \\ 0, & \Delta v_{x_log} \geq 0 \end{cases} \end{aligned} \tag{13}$$

The lateral driving safety cost is related to the relative distance and relative speed with the vehicle RV in the adjacent lane, and is expressed as:

$$\begin{aligned} C_{safe_lat} &= k_{v_lat}\lambda_{lat}\left(\Delta v_{x_lat}\right)^2 + k_{s_lat}/\left[\left(\Delta s_{x_lat}\right)^2+\varepsilon\right] \\ \Delta v_{x_lat} &= v_x^{RV} - v_x^{EV} \\ \Delta s_{x_lat} &= \sqrt{\left(x^{RV}-x^{EV}\right)^2+\left(y^{RV}-y^{EV}\right)^2} - l_v \\ \lambda_{lat} &= \begin{cases} 1, & \Delta v_{x_lat} < 0 \\ 0, & \Delta v_{x_lat} \geq 0 \end{cases} \end{aligned} \tag{14}$$

where v_x^{EV} and v_x^{RV} represent the longitudinal speeds of the EV and RV, respectively. $\left(x^{RV}, y^{RV}\right)$ and $\left(x^{RV}, y^{RV}\right)$ are the positions of the RV and EV, respectively. k_{v_lat} and k_{s_lat} are the weighting coefficients. ε is a very small value to prevent division by zero in the calculations. l_v is the safety factor that takes the vehicle length into account.

The ride comfort cost C_{com}^{EV} is directly related to the vehicle's acceleration and acceleration increment, and is defined as follows:

$$C_{com}^{EV} = \left(a^{EV}\right)^2 + \left(\Delta a^{EV}\right)^2 \tag{15}$$

In the equation, Δa^{EV} represents the vehicle's acceleration increment.

The driving efficiency cost C_{eff}^{EV} is defined as:

$$C_{eff}^{EV} = \left(v^{EV} - v_{desire}\right)^2 \tag{16}$$

where v_{desire} represents the vehicle's desired speed.

By combining the above cost functions, the integrated cost function for the lane-changing game of the intelligent autonomous vehicle is obtained:

$$C_{EV} = \omega_{safe}^{EV} C_{safe}^{EV} + \omega_{com}^{EV} C_{Ccom}^{EV} + \omega_{eff}^{EV} C_{eff}^{EV} \tag{17}$$

Where C_{safe}^{EV}, C_{com}^{EV}, and C_{eff}^{EV} represent the costs of driving safety, ride comfort, and driving efficiency, respectively. ω_{safe}^{EV}, ω_{com}^{EV}, and ω_{eff}^{EV} are the corresponding weight coefficients.

D. PREDICTION BASED STACKELBERG GAME MODEL

A multi-vehicle game framework for the human-robot hybrid driving environment based on surrounding vehicle trajectory prediction is established, as shown in Figure 4. This framework includes the leader-follower game between the autonomous intelligent vehicle and the human-driven vehicle, as well as a static game between the autonomous intelligent vehicle and other autonomous vehicles.

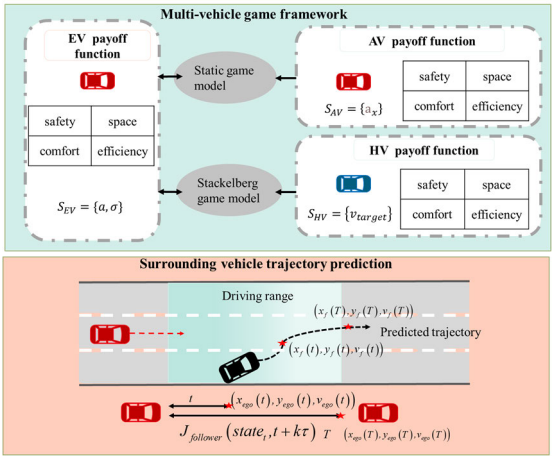


FIGURE 4. Trajectory prediction-based game-theoretic framework for multi-vehicle systems.

In calculating the cost function for the autonomous intelligent vehicle (EV), the predicted trajectory values of surrounding vehicles are introduced to calculate the predicted cost at future moments. This helps improve the robustness and safety of the leader-follower game decision model. The cost function of the EV, based on trajectory prediction information, is expressed as follows:

$$\begin{aligned} C_{leader}(t) &= C_{leader}\left(state_t^{EV}, state_{t+k\tau}^{HV}\right) \\ &= \omega_{safe}^{EV} C_{safe}^{EV}\left(state_t^{EV}, state_{t+k\tau}^{HV}\right) \\ &\quad + \omega_{com}^{EV} C_{com}^{EV}\left(state_t^{EV}\right) \\ &\quad + \omega_{eff}^{EV} C_{eff}^{EV}\left(state_t^{EV}\right) \end{aligned} \tag{18}$$

where $C_{leader}(t)$ represents the total cost function of EV at time t. $C_{leader}\left(state_t^{EV}, state_{t+k\tau}^{HV}\right)$ is the total predicted cost; $C_{safe}^{EV}\left(state_t^{EV}, state_{t+k\tau}^{HV}\right)$ is the safety prediction cost; $C_{com}^{EV}\left(state_t^{EV}\right)$ is the comfort prediction benefit; $C_{eff}^{EV}\left(state_t^{EV}\right)$ is the efficiency prediction benefit; τ is the prediction time interval; k is the number of predicted time frames; $state_t^{EV}$ is the state information of EV at time t; $state_{t+k\tau}^{HV}$ is the state information of HV at time $t + k\tau$. The safety cost function of EV is related to the state information of EV and HV, with the predicted state information of HV replacing the current state values.

When calculating the benefit function of surrounding human-driven vehicles, the predicted trajectory values are used to calculate the benefits at future moments, improving the robustness and safety of the decision-making model. The benefit function of human-driven vehicles, based on surrounding vehicle trajectory prediction information,

is expressed as follows:

$$
\begin{aligned}
&J_{follower}(t) \\
&\quad = \sum_{k=1}^{H} \xi^{k-1} J_{follower}(state_t, t+k\tau), \xi \in [0,1] \\
&J_{follower}(state_t, t+k\tau) \\
&\quad = J_{HV}(state_t, t+k\tau) \\
&\quad = \omega_{safe}^{HV} J_{safe}^{HV}(state_t, t+k\tau) \\
&\qquad + \omega_{space}^{HV} J_{space}^{HV}(state_t, t+k\tau) \\
&\qquad + \omega_{com}^{HV} J_{com}^{HV}(state_t, t+k\tau) \\
&\qquad + \omega_{eff}^{EV} J_{eff}^{HV}(state_t, t+k\tau)
\end{aligned} \tag{19}
$$

where $J_{follower}(t)$ represents the total benefit function of the game participants at time t; $J_{follower}(state_t, t+k\tau)$ is the total predicted benefit; $J_{safe}^{HV}(state_t, t+k\tau)$ is the safety prediction benefit; $J_{space}^{HV}(state_t, t+k\tau)$ is the lane change space prediction benefit; $J_{com}^{HV}(state_t, t+k\tau)$ is the comfort prediction benefit; $J_{eff}^{HV}(state_t, t+k\tau)$ is the efficiency prediction benefit; $state_t$ is the state of the vehicle at time t; ξ is the attenuation factor, which causes the contribution of future benefits to $J_{follower}(t)$ to decrease as the prediction time increases.

V. Stackelberg GAME-BASED Bi-LEVEL OPTIMIZATION METHOD FOR LANE-CHANGE DECISION

The Stackelberg game-theoretic model for lane-changing in mixed driving scenarios is equivalent to a bi-level optimization framework as shown in Figure 5, where the upper-level model employs model predictive optimization to establish a multi-objective function incorporating driving safety, traffic efficiency, and ride comfort, solving the autonomous vehicle's acceleration sequences to determine the lane-changing decision cost function and obtain optimal strategies, while the lower-level model utilizes genetic optimization to derive human-driven vehicles' optimal target speeds based on the EV's lane-changing strategy and feeds back the optimal target speed and vehicle state information to the upper-level model, ensuring that all decisions represent optimal strategies for both game participants under current driving conditions.

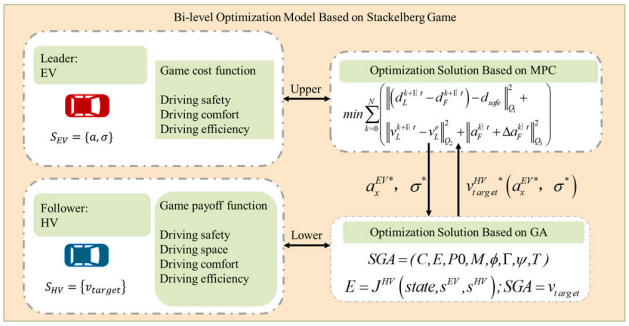


FIGURE 5. Stackelberg game-driven lane-change optimization model.

The bi-level optimization computational workflow for Stackelberg game-theoretic lane-changing decisions is detailed in Algorithm 1, where the payoff function of human-driven vehicles (HVs) serves as the fitness function in the genetic algorithm to derive target speeds for HVs with varying driving styles, which are then combined with HV state information as inputs to the upper-level optimization model. In the upper-level model predictive control (MPC) framework, the autonomous vehicle's decision variables include acceleration and lane-changing indicators: first, the optimal acceleration control strategy $\left\{a^{*,0|t}, \cdots, a^{*,k|t}, \cdots, a^{*,N-1|t}\right\}$ is resolved via MPC; subsequently, the comprehensive cost of lane-changing timing $t \sim t+N-1$ is calculated using Eq. (19) to select the timing with minimal cost, where $\mathcal{N}^o$ represents the game-theoretic cost value for the adjacent target lane and $\mathcal{N}^r$ for the ego lane, followed by iterative updates to the lane-changing timing $\tau_L^* = t + \tau$ and the EV's strategy set.

Algorithm 1

1: ***while*** $i < i_{max}$ ***do***

2: Initialize the genetic algorithm iteration count i_{max}, and generate the initial population of follower vehicles

3: Calculate individual fitness values according to the follower vehicle's profit function

4: Perform selection, crossover, and mutation operations to evolve the population.

5: Obtain the optimal population signal s_F^* for follower vehicles. Derive the follower's initial state values from s_F^* and transmit them to the leader vehicle.

6: Initialize the lane-change timing τ_L^*, $\tau_{F\in\mathcal{N}}^* = 0$

7: $$min \sum_{k=0}^{N} \left(\begin{array}{c} \left\| \left(d_L^{k+1|t} - d_F^{k+1|t}\right) - d_{safe} \right\|_{Q_1}^2 + \\ \left\| v_L^{k+1|t} - v_L^e \right\|_{Q_2}^2 + \left\| a_F^{k|t} + \Delta a_F^{k|t} \right\|_{Q_3}^2 \end{array} \right)$$

Generate the control policy sequence $a_F^{*,0|t} \sim a_F^{*,N-1|t}$

8: Calculate the ego vehicle's original lane cost $C_{F\in\mathcal{N}^o}^{0|t} \sim C_{F\in\mathcal{N}^o}^{N-1|t}$ and lane-change game cost $C_{F\in\mathcal{N}^r}^{0|t} \sim C_{F\in\mathcal{N}^r}^{N-1|t}$ based on the game-theoretic cost expression (19) and $a_F^{*,0|t} \sim a_F^{*,N-1|t}$.

9: ***If*** $min\left(\sum_{k=t}^{k=\tau} C_{L\in\mathcal{N}^o}^{\tau|t} + \sum_{k=\tau}^{k=t+N-1} C_{L\in\mathcal{N}^r}^{\tau|t}\right)$ ***then***

10: Lane-change timing $\tau_L^* = t + \tau$

11: ***end***

12: Update the policy set $S^* = \left\{s_L^*, s_F^*\right\}$

13: ***end***

VI. SIMULATION

A joint simulation model of Prescan-Carsim-Simulink has been established. As shown in Figure 6, the main-follower game lane change is modeled and simulated under three typical mixed driving scenarios. The decision variables a_x^{EV*} and σ^* of the autonomous intelligent vehicle, calculated from the simulation, are transmitted to the trajectory planning

module to find the optimal trajectory. Finally, the planned trajectory information is passed to the lower-level motion control module.

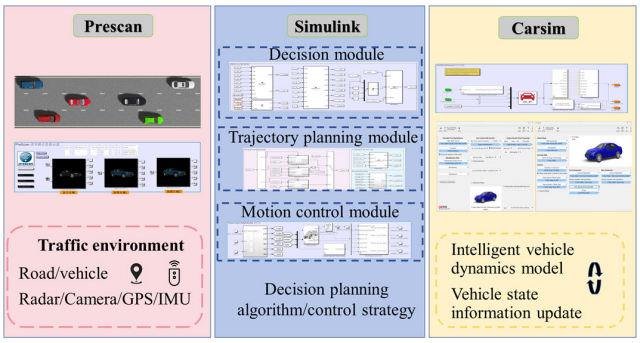


FIGURE 6. Co-simulation schematic for lane-change decision-making.

A. LANE-CHANGE SCENARIO SELECTION

For the typical free lane change scenario shown in Figure 7, the game process between the human-driven vehicle RV1 with different driving styles and the autonomous intelligent vehicle EV is simulated and analyzed. RV1, depending on its driving style, will have different target speeds to maximize its own rewards. EV uses the main-follower game-based lane change decision method to make optimal decisions based on the different driving behaviors of RV1.

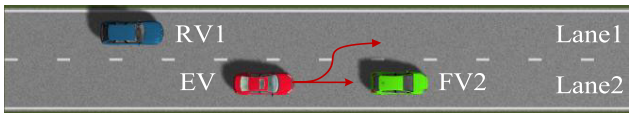


FIGURE 7. Free lane-change scenarios with diverse driving styles.

Table 1 provides the weight coefficients of the reward functions for three driving styles: aggressive, normal, and conservative.

TABLE 1. Different driving styles correspond to different weight parameters.

Driving styles	Aggressive	Normal	Conservative
ω_{safe}^{HV}	0.5	1	2
ω_{space}^{HV}	0.5	1	2
ω_{com}^{HV}	0.1	0.3	0.5
ω_{eff}^{EV}	0.8	0.2	0.1

At the start time, EV is traveling at a speed of 20 m/s in lane 2, with the vehicle FV2 in lane 2 traveling at a speed of 14 m/s, 20 meters ahead of EV. The target lane vehicle RV1 is traveling at a speed of 15 m/s, positioned 5 meters behind EV.

When EV signals its intention to change lanes by turning on the turn signal, RV1 reacts differently depending on its driving style, either decelerating, accelerating, or making other speed adjustments during the lane change process. The lane change decision planning results are shown in Figure 7.

As seen in Figure 8(a), the trajectory curves are marked with discrete points at 2-second intervals to show the relative positions of different vehicles at the same moment. From Figure 8(b), it can be observed that to maintain a safe distance from FV2 and RV1, EV can adjust its speed in real-time to approach the target speed, depending on RV1's driving style. In Figure 8(c), it is evident that the driving cost for EV in its original lane is lower than in the target lane before the lane change starts. After the lane change begins and until it is completed, the driving cost in the target lane is lower than in the original lane.

In Figure 8(d), when the lane change flag is set to 1, EV is cruising in the original lane; when the lane change flag is set to 2, EV is changing lanes; when the lane change flag is set to 3, EV has completed the lane change and is cruising in the target lane. When RV1 is conservative, EV starts changing lanes at t = 0.2s and completes the lane change at t = 2.05s. When RV1 is of the general type, EV starts changing lanes at t = 1s and completes the lane change at t = 2.9s. When RV1 is aggressive, EV starts changing lanes at t = 1.45s and completes the lane change at t = 3.35s. From Figure 8(d), the distance between EV and RV1, as well as between EV and FV2, shows that when the lane change begins, the distance between EV and RV1 increases over time according to RV1's driving style. After the lane change, the minimum distance between EV and the general RV1 is 10.43m, and the minimum distance with the aggressive RV1 is 8.5m. At the beginning of the lane change, the distance between EV and FV2 decreases over time, depending on RV1's driving style. When RV1 is conservative, the minimum distance between EV and FV2 is 10.15m; when RV1 is of the general type, the minimum distance is 6.88m; and when RV1 is aggressive, the minimum distance is 5.94m.

From Figure 8(e), it can be seen that when EV begins changing lanes, the conservative RV1 starts to decelerate from an initial speed of 15m/s to 14m/s. After EV and RV1 are in the same lane, RV1 gradually accelerates. The general RV1 accelerates from 15m/s to 18.8m/s when the lane change starts, while EV's speed is 19.5m/s. Afterward, RV1 begins to decelerate slowly to 18m/s. The aggressive RV1 has a tendency to accelerate from the beginning, and by the time of the lane change start at t = 1.45s, its speed has increased to 17m/s. By the time the lane change is completed, the speed reaches 20.2m/s, while EV's speed is 19.5m/s, after which EV starts to accelerate slowly to 21m/s, while RV1's average speed remains at 20.5m/s.

The simulation results show that in the lane change interaction scenarios with conservative, general, and aggressive human driving vehicles, EV can reasonably adjust its lane change strategy based on the different driving styles,

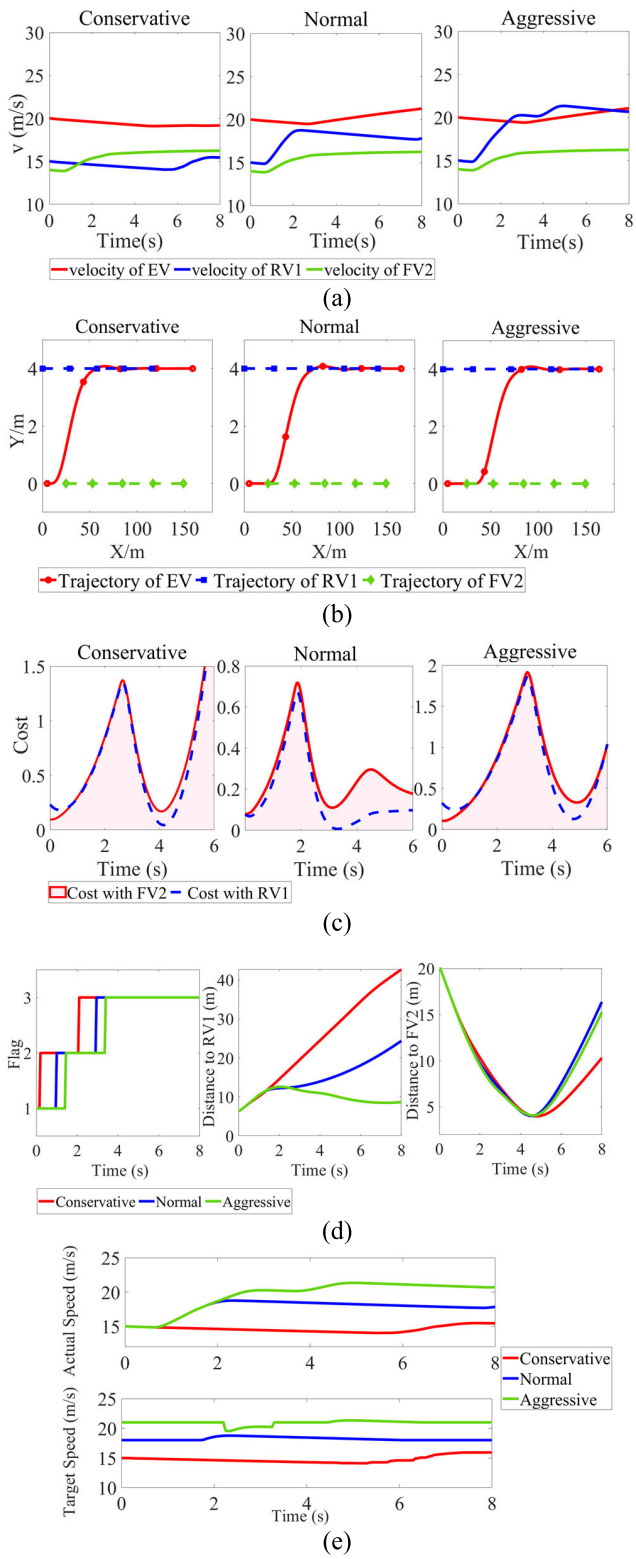


FIGURE 8. Results in lane-change scenario.

achieving smooth trajectory planning, and demonstrating high safety in speed and distance control.

B. MANDATORY LANE CHANGE SCENARIO

As shown in Figure 9, when FV3 from lane 3 inserts in front of EV2, entering Vehicle 2, the forced lane change decision-making planning results for EV, based on the trajectory prediction of FV3, are shown in Figure 9.

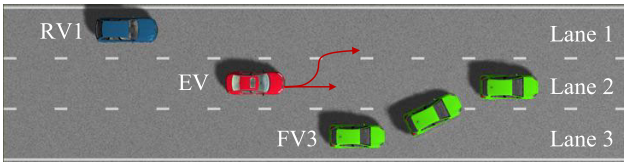


FIGURE 9. Mandatory lane-change scenario due to intrusive Cut-in.

EV is driving at 20 m/s in lane 2, while FV3 in lane 3 is initially traveling at 15 m/s and is 18 meters ahead of EV. FV3 then quickly cuts into lane 2. At this moment, RV1 is driving at 15 m/s and is 5 meters behind EV in the longitudinal position.

From Figures 10(a) to (d), it can be seen that without the FV3 trajectory prediction, EV starts an emergency lane change at t = 2.6s and completes the lane change rapidly by t = 3.55s, with the lane change trajectory showing significant fluctuations. When FV3's trajectory prediction information is available, EV begins the lane change earlier at t = 1.5s and completes it smoothly by t = 3.45s, with a smoother trajectory. Figures (f) to (i) show the trajectories of EV, FV3, and RV1 at different times, along with the predicted trajectories of FV3 and RV1.

From Figures 10(j) and (k), it can be seen that without the FV3 trajectory prediction, the lane change only begins when the distance between EV and FV3 is 6.41m, and during the lane change, the minimum distance approaches 3.68m. With the FV3 trajectory prediction, EV begins the lane change when the distance to FV3 is 11.61m, and the minimum distance during the lane change is 4.81m. From Figures 10(l) and (m), it is evident that without FV3 trajectory prediction, the lane change starts at t = 2.6s; with FV3 trajectory prediction, the lane change starts at t = 1.5s, and EV's cost of driving in the target lane is lower than that in the original lane, enabling the lane change. From Figures 10(n) and (o), when FV3's trajectory is predicted, the lane change timing is optimal. Compared to when there is no FV3 trajectory prediction, the lane change time is longer, and the heading angle change is reduced to 0.16rad, a 60% reduction; the yaw rate change is reduced to 0.47rad/s, a 67.59% reduction, allowing for a smoother lane change.

Simulation results show that with FV3 trajectory prediction, EV can anticipate the driving intentions and trajectories of surrounding vehicles, choose the optimal lane change timing, and ensure that there are no sharp turns during the entire lane change process. This leads to a smoother lane change, preventing potential collisions with vehicles suddenly entering from the adjacent lane, thereby improving lane change safety.

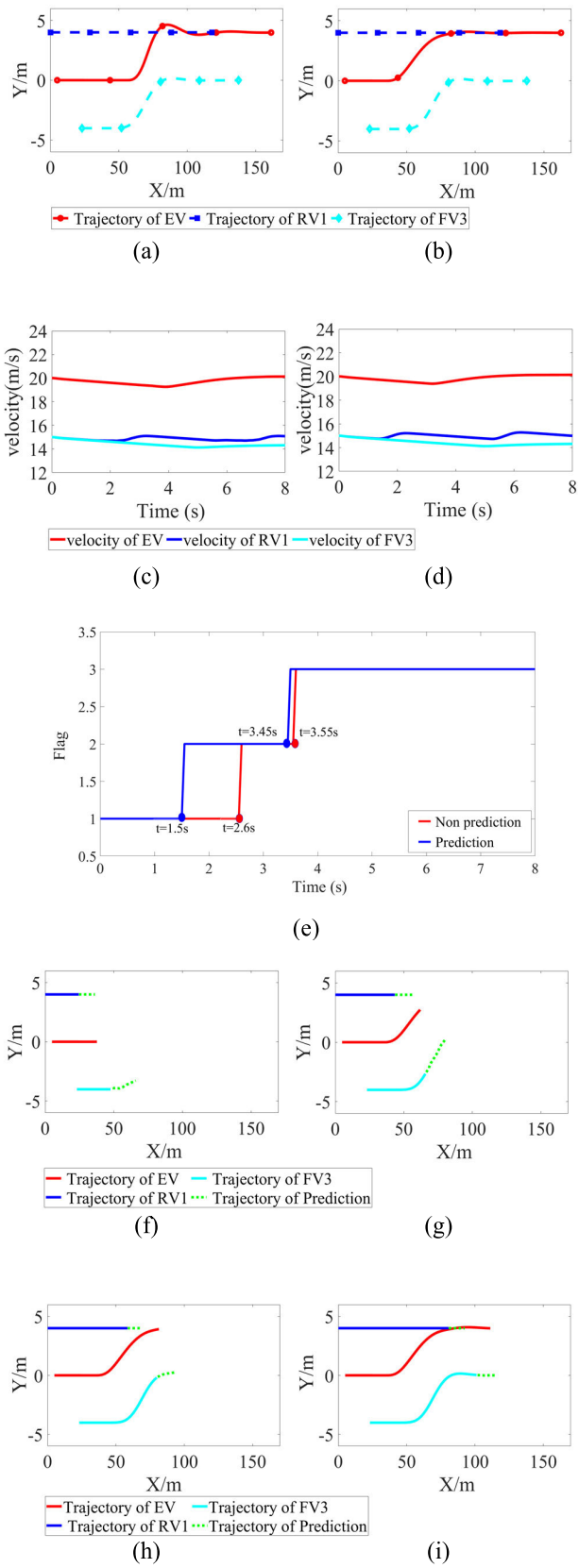


FIGURE 10. Results in mandatory lane-change scenario.

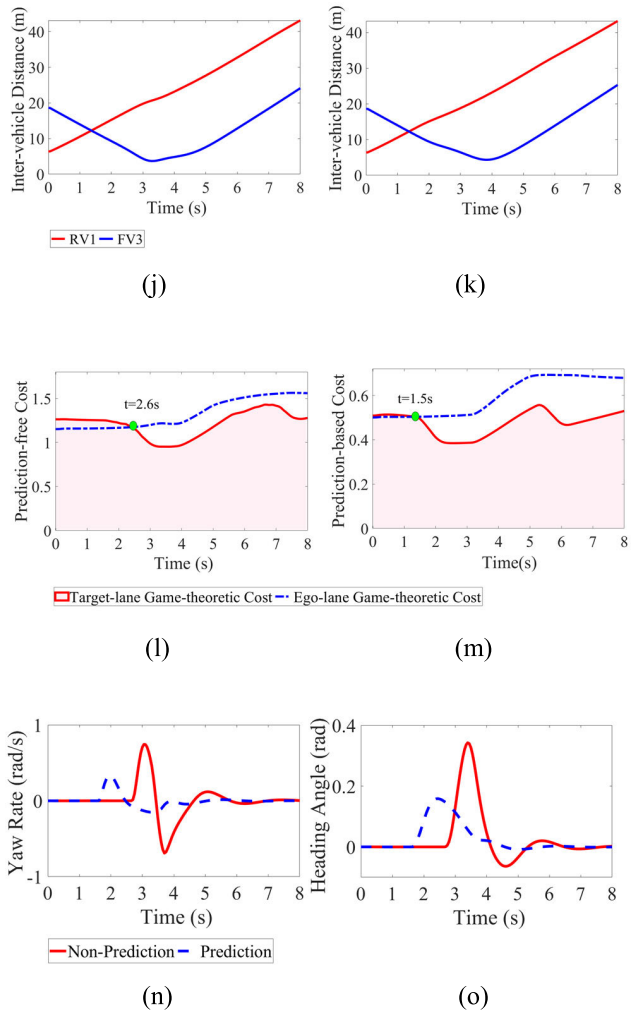


FIGURE 10. *(Continued.)* Results in mandatory lane-change scenario.

C. MULTI-LANE SELECTION SCENARIO

As shown in Figure 11, the lane change game process in the mixed-driving traffic scenario is analyzed.

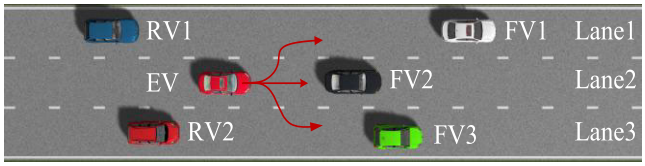


FIGURE 11. Multi-lane free lane-change scenarios with dynamic selection.

To determine whether the EV can choose the optimal time for a smooth lane change, the simulation results of the leader-follower game lane change decision are shown in Figure 11. The initial state parameters of each vehicle are set in Prescan, as shown in Table 2.

From Figures 12(a) to (d), it can be seen that since the EV starts with a higher speed in lane 1 while the leading vehicle FV2 has a lower speed, the risk of traveling in lane 2 gradually

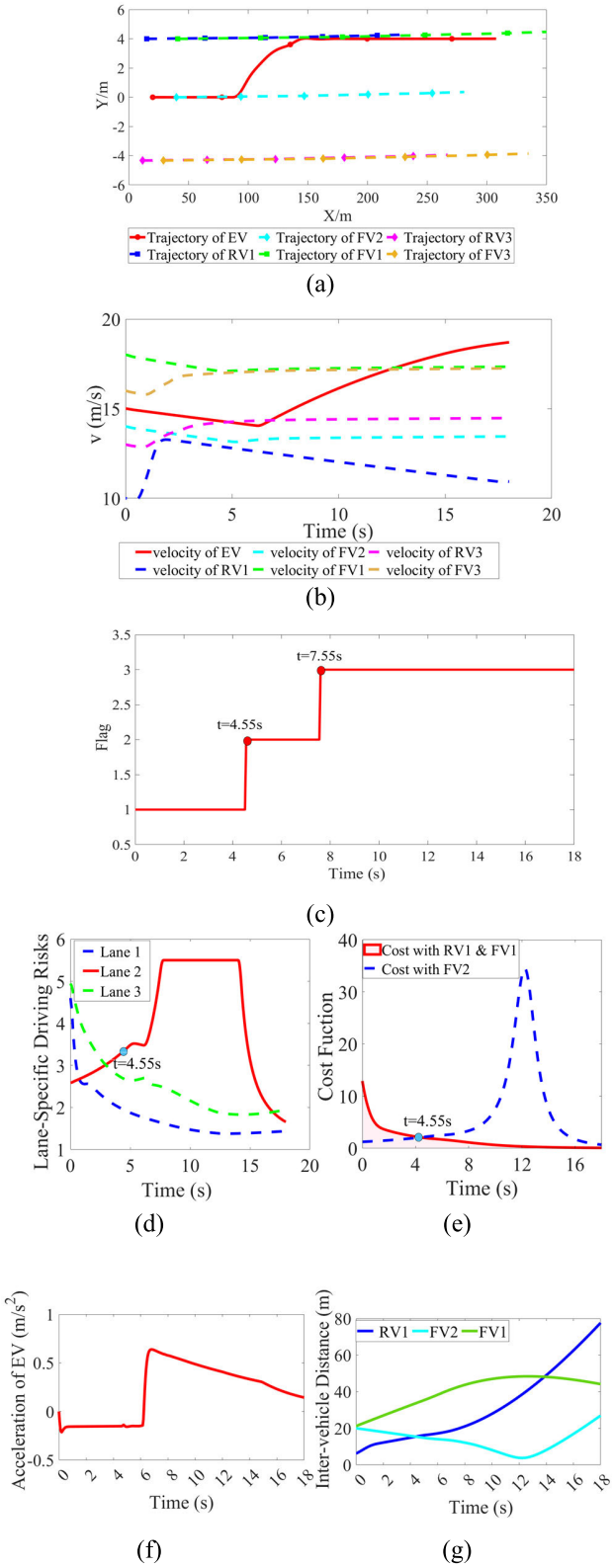


FIGURE 12. Results in multi-lane free lane-change scenario.

increases until it reaches the set risk threshold of 5.5. At this point, the risk of traveling in lane 1 starts to be lower than in lane 3, prompting the EV to choose lane 1 for the lane change. In Figure 12(e), from t = 4.55s, the EV's travel cost in the target lane is lower than in the original lane, which signals the optimal moment for the lane change. The EV continues to change lanes smoothly until t = 7.55s, when the lane change is completed. In Figure 12(g), due to the EV's initial speed being higher than FV2's, the distance between the EV and FV2 gradually decreases. When the distance reaches 15.45m, the EV begins the lane change. RV1 first accelerates, but upon seeing the EV changing lanes, it starts to decelerate, causing the distance between EV and RV1 to gradually increase. The EV accelerates after completing the lane change to achieve a higher travel speed, which causes its distance from FV1 to gradually decrease.

TABLE 2. Initial parameters for multi-lane selection scenario.

Vehicle	EV	RV1	FV1	FV2	RV3	FV3
Y/m	20	15	41	40	12	30
X/m	0	4	4	0	-4	-4
Initial speed/m·s^{-1}	15	10	18	14	13	16

VII. CONCLUSION

(1) To address lane-changing conflicts between autonomous and human-driven vehicles in mixed traffic environments, a Stackelberg game framework was established between autonomous vehicles (leaders) and surrounding human-driven vehicles (followers) across three typical lane-changing scenarios. An enhanced Stackelberg game-based lane-changing decision-making model incorporating human driver trajectory prediction was developed to improve both safety and operational efficiency during lane transitions.

(2) Considering safety, comfort, and efficiency requirements, a comprehensive utility function accounting for different driving styles was formulated for human-driven vehicles, while a multi-objective cost function quantifying acceleration variations and traffic efficiency was established for autonomous vehicles. This dual-function architecture ensures rational decision-making under diverse driving conditions.

(3) For solving the Stackelberg game optimization problem, a bi-level optimization method integrating model predictive control (MPC) and genetic algorithms (GA) was proposed to jointly optimize lane-changing acceleration sequences and timing parameters. Prescan-Carsim-Matlab/Simulink co-simulation results demonstrate significant improvements over non-game-theoretic strategies: 60% reduction in heading angle variation and 67.59% decrease in yaw rate during lane changes, effectively

enhancing safety, comfort, and traffic efficiency. These findings validate the feasibility and rationality of the proposed methodology.

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BAICHUAN SHI received the B.S. degree in engineering mechanics from Beijing Institute of Technology, Beijing, China, in 2024, where he is currently pursuing the M.S. degree in mechanical engineering. His research interests include vehicle system dynamics and control and autonomous driving decision making.

LI ZHAI (Member, IEEE) received the Ph.D. degree in vehicle engineering from Beijing Institute of Technology, Beijing, China, in 2004. From 2009 to 2019, she was an Associate Professor. From 2013 to 2014, she was a Visitor Scholar in electrical and computer engineering with the University of Missouri-Rolla, Rolla, MO, USA. Since 2020, she has been a Professor with the National Engineering Laboratory for Electric Vehicle and the School of Mechanical Engineering, Beijing Institute of Technology. Her research interests include the include dynamics control, motor drive system control, and EMC of EVs

CHANG LIU received the M.S. degree in automotive engineering from Beijing Institute of Technology, Beijing, China, in 2024. Her research interests include vehicle system dynamics and control and vehicle path tracking control.